A simple wing from design to STL

MIE 446 Aerospace Structures

Build one small tapered semi-wing in Wing Lab, transfer the design to Colab, and generate one connected printable part. This worked example is a non-flying fabrication demonstrator.

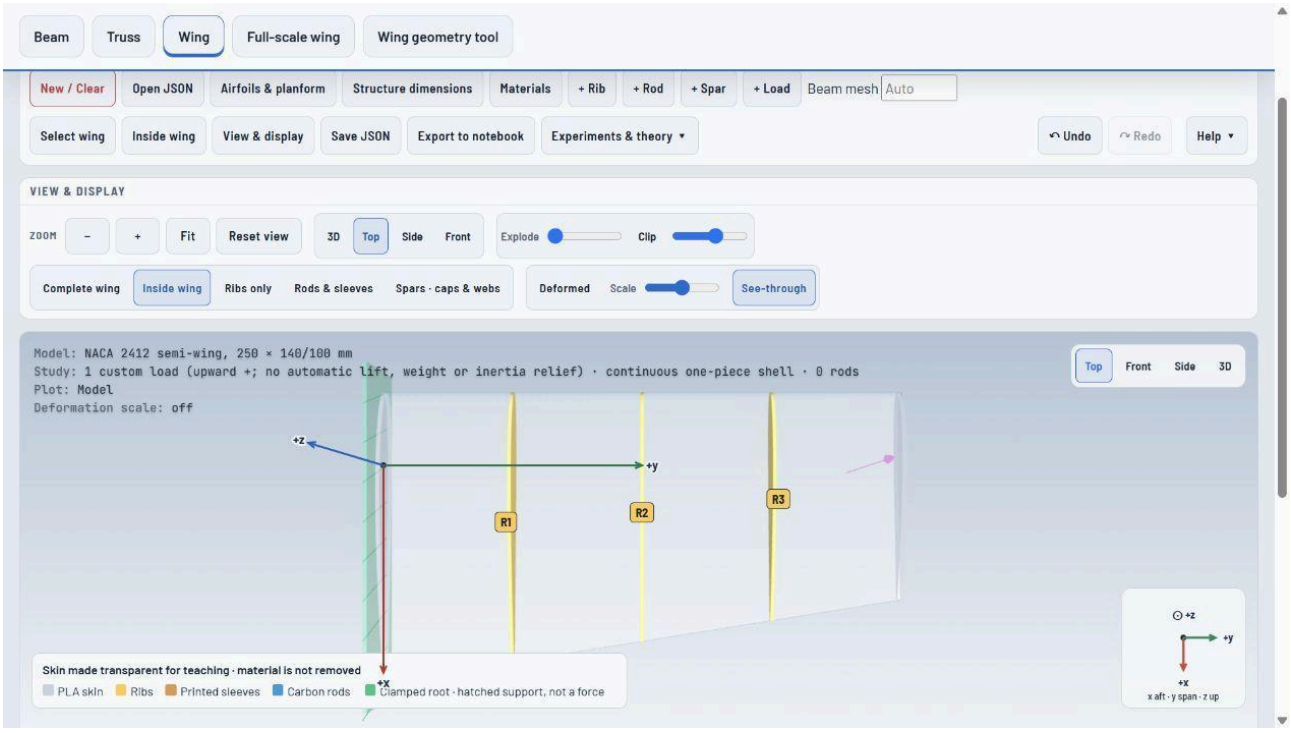
Open Wing Lab

Open the course notebook in Colab

The example you will build

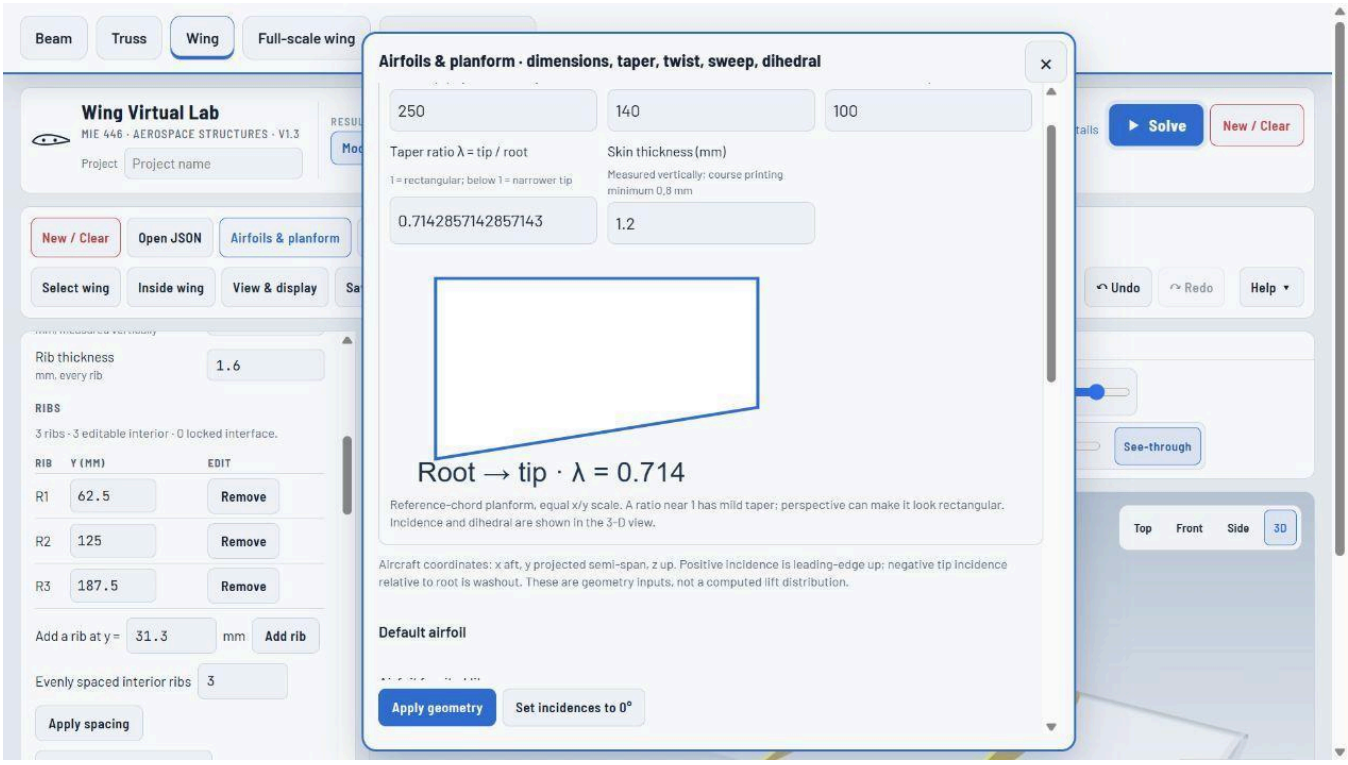
Input	Value
Airfoil	NACA 2412
Semi-span and root and tip chords	250 mm; 140 mm; 100 mm
Skin and rib thickness	1.2 mm; 1.6 mm
Structure	1 module; 3 interior ribs; no external rods or bonded spars
Sweep and dihedral and incidence	All 0 degrees
Virtual point load	+0.1 N at y = 250 mm and x/c = 0.30
Printer envelope	300 × 300 × 300 mm

Design basis: make a compact one-piece teaching model that fits the course printer. Use a small prescribed tip load to inspect the fixed-root beam response. This load is an illustrative input, not calculated aircraft lift. Do not physically load or fly the print.



1 Set the geometry and ribs

1. Open Wing Lab and use New / Clear to start a new study. Save your previous study first if you need it.
2. Open Airfoils & planform. Choose NACA 2412 and enter semi-span 250 mm, root chord 140 mm and tip chord 100 mm. Keep sweep, dihedral and incidence at zero. Set skin thickness to 1.2 mm and Apply geometry. For sweep, dihedral or twist, use the optional page that follows.
3. Use one printed module. In Structure dimensions set rib thickness to 1.6 mm and use 3 evenly spaced interior ribs. Their stations are 62.5, 125 and 187.5 mm. Leave rods and bonded spars empty.

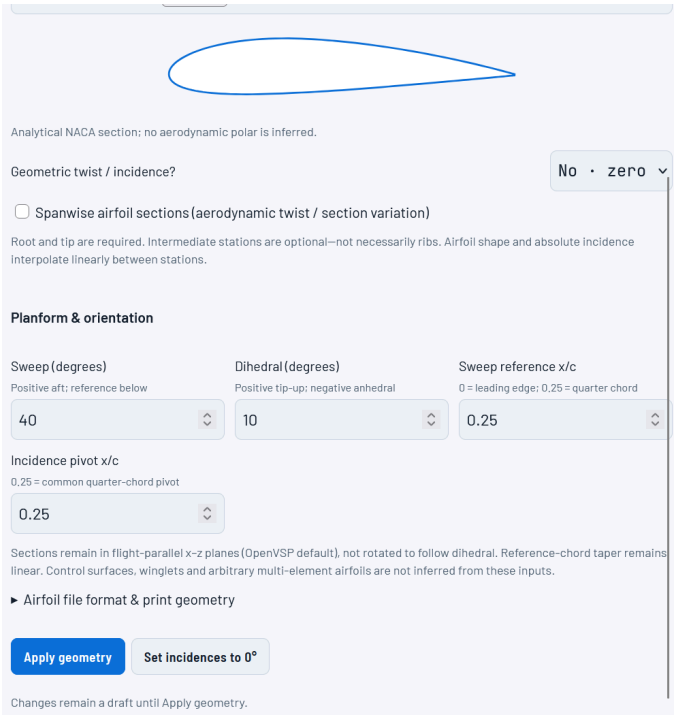


The taper ratio is tip chord / root chord = 100 / 140 = 0.714. The displayed transparent skin is a viewing aid; it does not remove material from the printable model.

Optional sweep dihedral and twist

Use this page if you want a more complex wing. Keep the simple example unchanged until you have saved its editable JSON.

Open Airfoils & planform. Under Planform & orientation, enter Sweep and Dihedral in degrees: positive sweep moves the tip aft; positive dihedral raises the tip. Negative dihedral lowers the tip.



Airfoils & planform controls. The screenshot values are examples, not required settings.

Sweep reference x/c and Incidence pivot x/c are chord fractions, not angles. Set both to 0.25 for a quarter-chord reference. A sweep reference of 0 uses the leading edge.

For geometric twist, change Geometric twist / incidence? to Yes and enable Spanwise airfoil sections. At root eta = 0 and tip eta = 1, enter the absolute incidence angles; positive incidence raises the leading edge. Root +2 degrees and tip 0 degrees give 2 degrees of washout. Use Add intermediate station for another section, such as eta = 0.5.

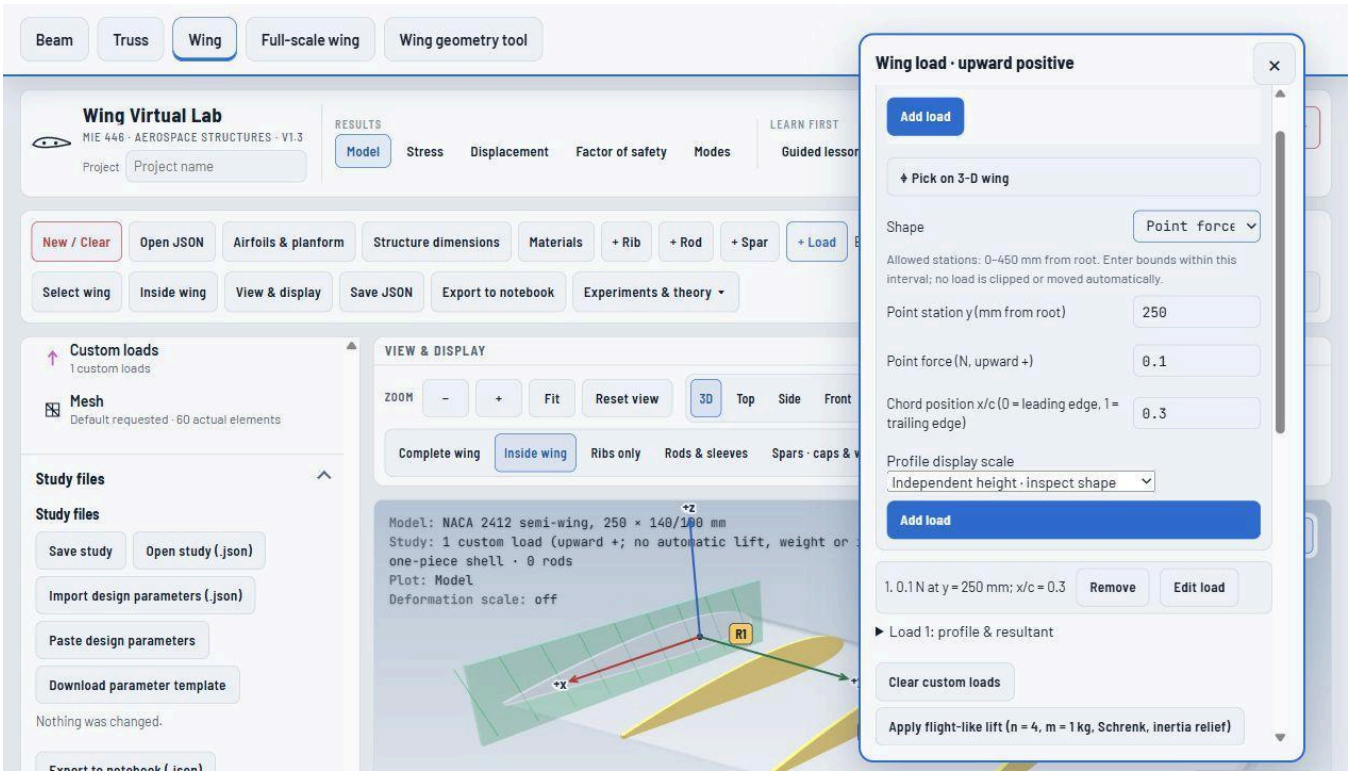
To vary the airfoil along the span, choose the profile separately at each station, from the library or a supported .dat / .txt file. Root and tip are required. The software interpolates airfoil shape and incidence between stations; stations do not automatically add ribs.

Press Apply geometry, then Solve. Inspect Top, Front and Side views and the analysis warnings. Keep the load case fixed when comparing designs; airfoil and incidence changes do not automatically calculate lift. The nonzero-angle analysis uses a small-displacement spatial beam model, not solid or shell FEA.

Save JSON and Export to notebook again. Import the new transfer file into the notebook and check the stations, angles, CAD, mass and 300 mm printer envelope before exporting STL. These changes can invalidate earlier build checks; thinner airfoils or excessive skin thickness can close the cavity.

2 Apply a load and solve

4. Clear any previous custom loads. Use + Load and select Point force.
5. Enter y = 250 mm, force = +0.1 N, and x/c = 0.30. Press Add load. Check that only this one load is listed.
6. Press Solve. Inspect the model and displacement plot. Keep the default mesh for this simple example; the completed run used 60 beam elements.



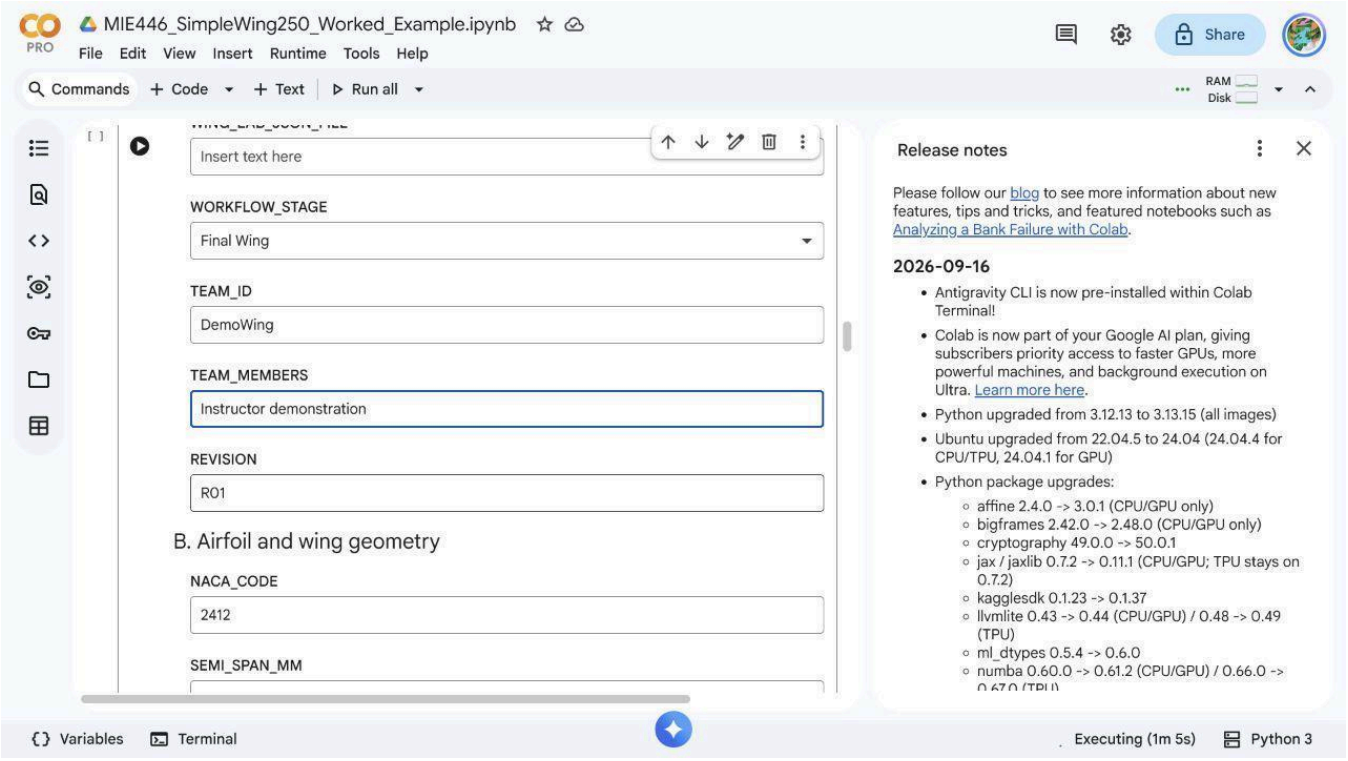
Hand check: root shear magnitude = 0.1 N. Root bending moment magnitude = 0.1 × 0.250 = 0.025 N m. The completed browser solve reported a maximum displacement of about 0.03 mm. This is a beam-model result, not a physical test or certification of the print.

Save two different files

Use Save JSON for an editable Wing Lab backup. Use Export to notebook for the transfer JSON. Keep both. To change the design later, open the editable backup in Wing Lab, solve again and export a new transfer file.

3 Prepare your notebook

7. Open the course Colab notebook and save a copy in Drive. Run the setup cell and wait for Design tool ready.
8. In STUDENT INPUT FORM choose DESIGN_SOURCE = Wing Lab JSON and WORKFLOW_STAGE = Final Wing. Enter your own team ID, team members and revision. Keep all printer dimensions at 300 mm.
9. Complete the engineering responses and AI declaration truthfully. Run the form cell after changing entries. In JSON mode, the uploaded file supplies the geometry; the baseline geometry values displayed in the form are not used.



Write your own explanation

State why you chose the airfoil, your load and its location, and what you checked. For this example, reducing the comparison tip chord from 100 to 90 mm reduces area and increases aspect ratio at fixed span. Explain the trend before running the comparison. Identify a print defect the code cannot detect, such as poor layer adhesion.

The screenshot uses DemoWing and Instructor demonstration. Replace those with your own team information. Do not copy a demonstration identity or claim a coupon test you did not perform.

4 Import and check the design

Upload only the transfer JSON from Export to notebook. The notebook must report Wing Lab → CAD import accepted and show NACA 2412, semi-span 250 mm, chords 140/100 mm, one module, three ribs and zero rods.

10. Open the Colab Files panel and use Upload to session storage to upload the transfer JSON.
11. Set WING_LAB_JSON_FILE to the exact uploaded filename or its /content/ path. Run the form cell, then the Import the Wing Lab design cell. Alternatively, leave the field empty and choose the file when the import cell asks.
12. Run Step 3, the automatic 2D analysis. Check the area, taper ratio and rib layout before building CAD.

Check the import message

The recorded notebook output was Wing Lab → CAD import accepted: NACA 2412; semi-span 250 mm; chords 140/100 mm; one module; three physical ribs; zero external rods and zero integral printed spars. Check these values against your own file. The source JSON SHA-256 begins 0b8bfd8c1b68f29a.

Expected checks: semi-wing area = 250 × (140 + 100) / 2 = 30,000 mm²; equivalent full-wing area = 60,000 mm²; aspect ratio = 500² / 60,000 = 4.1667. The physical print is one semi-wing, not the full-wing equivalent.

5 Build and validate the CAD

Run the coupon-stage cell. This example has no external rods, so the notebook reports that a rod-fit coupon is not applicable. If your own design includes rods, use your actual tested coupon record instead.

Run Build and inspect the 3D model. Wait for the report and require Overall result PASS before export. The completed example produced the values below.

CAD check	Completed example
Connected printable modules	1
Estimated fully dense PLA mass	109.72 g
Actual CAD bounding box	140.05 × 250.00 × 17.03 mm
Usable printer envelope	300 × 300 × 300 mm
Overall validation	PASS

Passing these checks confirms the tested CAD conditions. It does not certify layer bonding, local buckling, joint strength or flight safety. Printed mass can differ from the fully dense CAD estimate.

6 Export the files and open the STL

Run Create and download the final ZIP. The completed example reported MIE446_DemoWing_R01.zip with 11 hashed artifacts. Save your own ZIP and your completed notebook copy.

Keep the right file for each job

File	Use
Editable Wing Lab JSON	Reopen and edit the design in Wing Lab
Transfer JSON	Import geometry into the course notebook
Completed notebook IPYNB	Keep the form, code and recorded outputs
Module STL	Import the one-piece geometry into OrcaSlicer
Verified ZIP	Keep CAD formats, parameters, reports and hashes together

13. Extract the ZIP and locate the module STL. One module means one continuous part. Retain the STEP, 3MF, parameter records and validation files as well.
14. Open the module STL in OrcaSlicer using the staff-approved profile for the actual 300 mm printer, nozzle and PLA. Confirm millimetres and 100 percent scale.
15. Check orientation, bed contact, supports or brim clearance, and every layer. Ask staff to review the print before starting the printer.

STL contains geometry, not printer instructions. OrcaSlicer generates the machine-specific G-code after slicing. The screenshots show the actual Wing Lab model and Colab input form. Import and CAD results above are transcribed from the saved executed notebook. This guide does not show a completed slicer review or a physical print.